



Description

The data is noted, assuming a normal speed of the cycle and force & Pressure applied by the brakes on the disc.

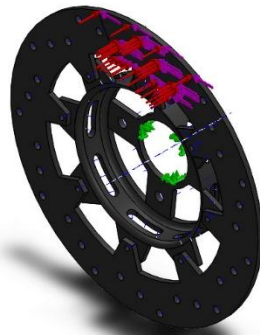
Simulation of Disc Brake

Date: 5 May 2025
Designer: Solidworks
Study name: Disc Brake Static
Analysis type: Static

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Model Information



Model name: kh
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line3 	Solid Body	Mass:10.646 kg Volume:0.00147861 m ³ Density:7,200 kg/m ³ Weight:104.33 N	C:\Users\prati\Downloads\discbrake\brake.SLDPRT May 27 11:10:49 2017
Cut-Extrude2 	Solid Body	Mass:6.89699 kg Volume:0.000957915 m ³ Density:7,200 kg/m ³ Weight:67.5905 N	C:\Users\prati\Downloads\discbrake\njjn.SLDPRT May 27 11:10:49 2017



Study Properties

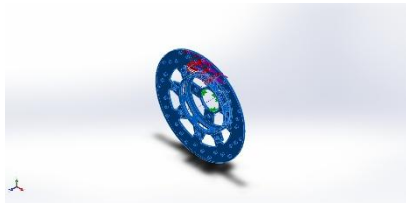
Study name	Disc Brake Static
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\prati\Downloads\discbrake)

Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

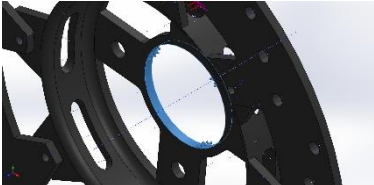


Material Properties

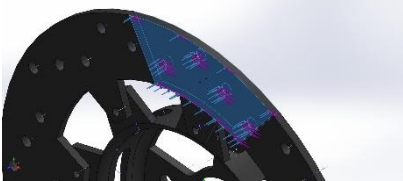
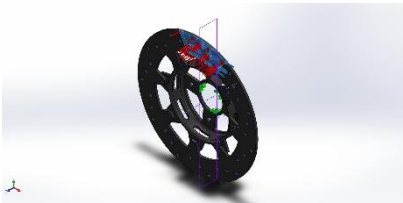
Model Reference	Properties	Components
	Name: Gray Cast Iron Model type: Linear Elastic Isotropic Default failure criterion: Mohr-Coulomb Stress Tensile strength: 1.51658e+08 N/m² Compressive strength: 5.72165e+08 N/m² Elastic modulus: 6.61781e+10 N/m² Poisson's ratio: 0.27 Mass density: 7,200 kg/m³ Shear modulus: 5e+10 N/m² Thermal expansion coefficient: 1.2e-05 /Kelvin	SolidBody 1(Split Line3)(brake-1), SolidBody 1(Cut-Extrude2)(njjn-1)
Curve Data:N/A		



Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	994	8.7738e-05	-6.10352e-05	994
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details			
Pressure-1		Entities:	2 face(s)		
		Type:	Normal to selected face		
		Value:	1e+06		
		Units:	N/m^2		
		Phase Angle:	0		
		Units:	deg		
Force-1		Entities:	2 face(s), 1 plane(s)		
		Reference:	Right Plane		
		Type:	Apply force		
		Values:	---, ---, -, ,497 N		



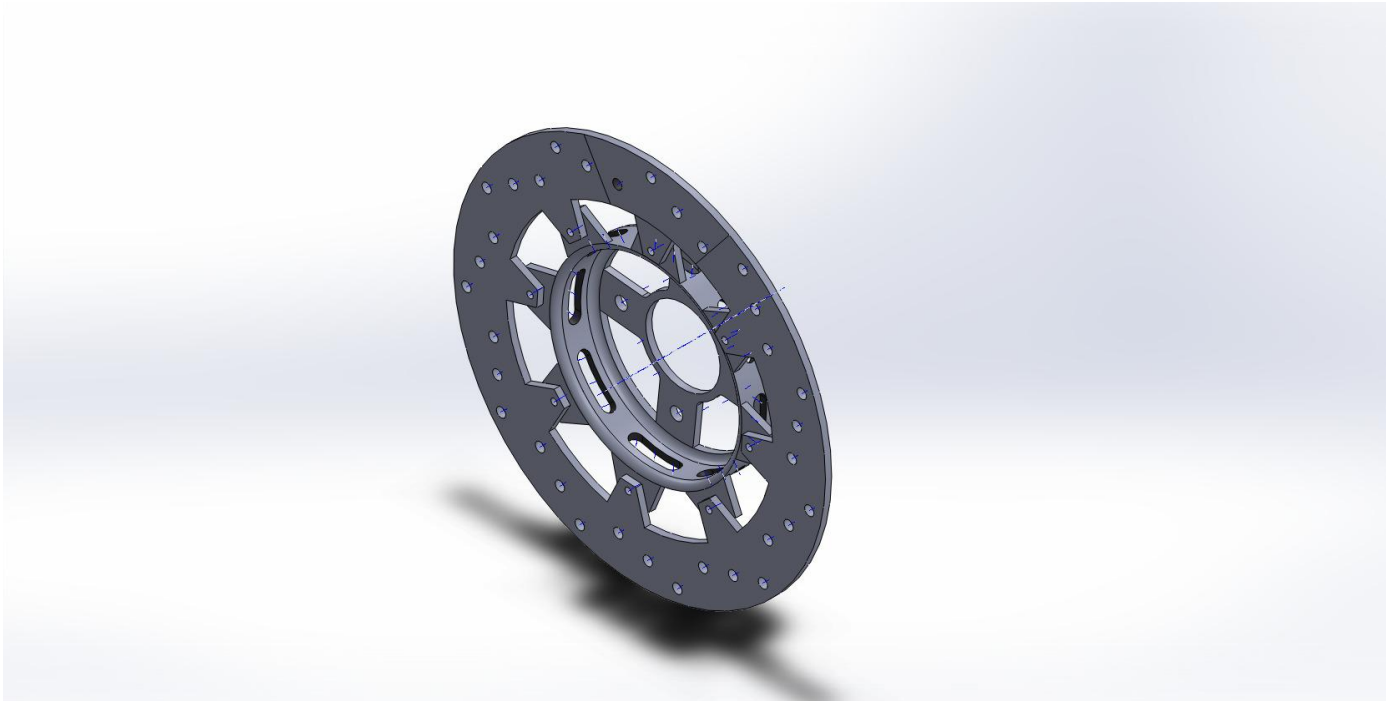


Image-1

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh



Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	30.8625 mm
Minimum element size	3.82683 mm
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

Mesh information - Details

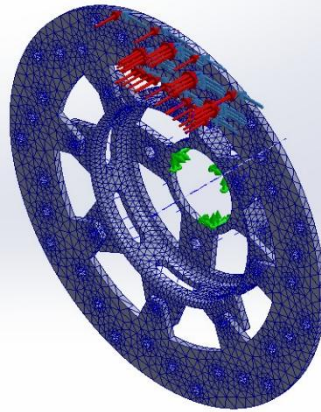
Total Nodes	44430
Total Elements	22231
Maximum Aspect Ratio	7.5851
% of elements with Aspect Ratio < 3	96.4
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:03
Computer name:	MSI

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Model name: kh
Study name: Disc Brake Static(-Default-)
Plot type: Mesh Quality1



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kh-Disc Brake Static-Quality-Quality1

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	994	8.7738e-05	-6.10352e-05	994

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

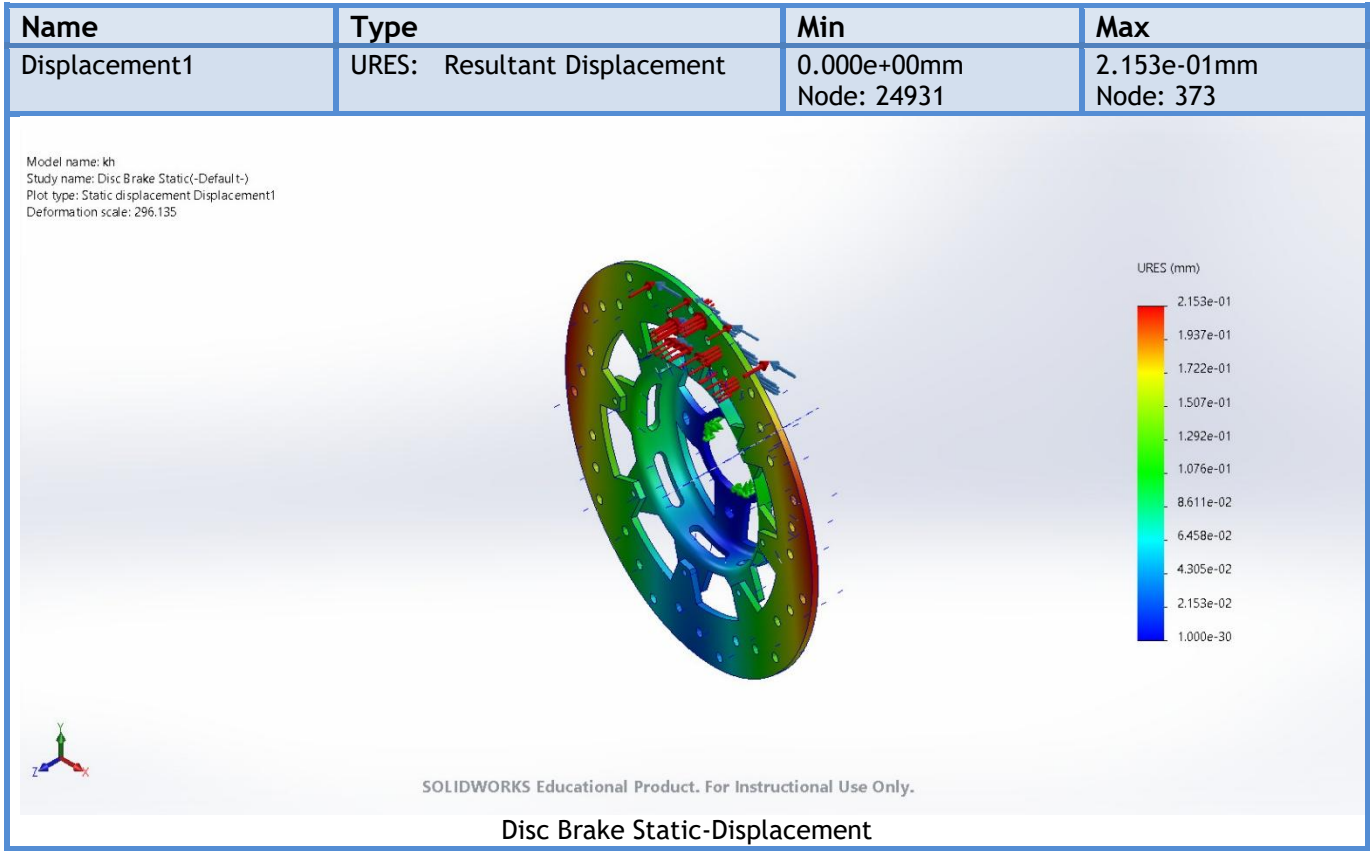
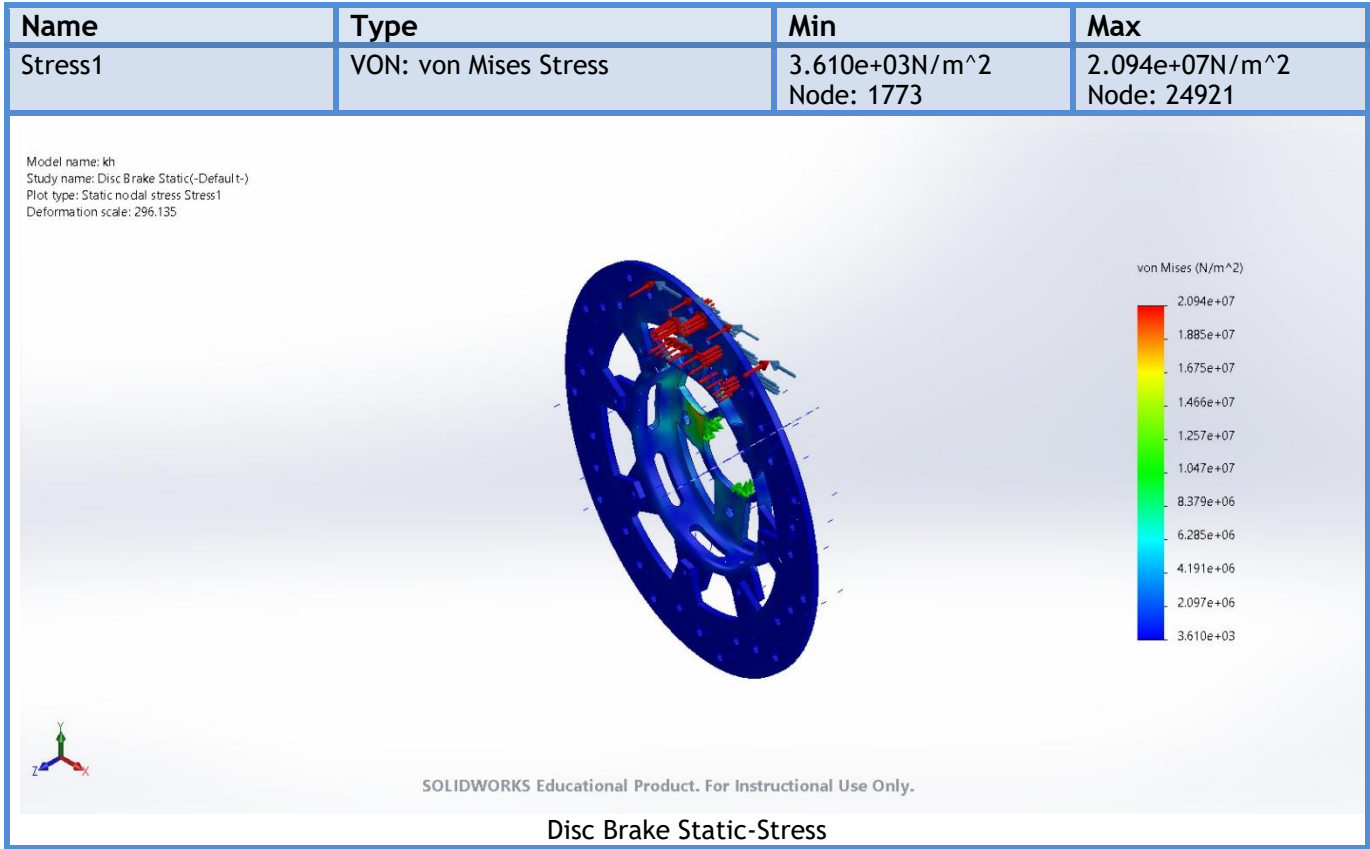
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.00064984	0.000137269	0.000370268	0.000760416

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

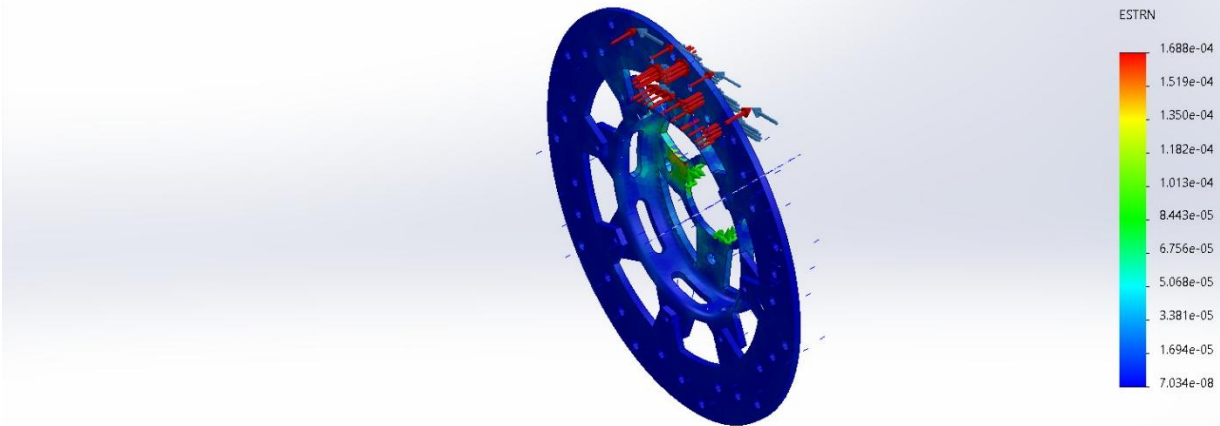


Study Results



Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	7.034e-08 Element: 1640	1.688e-04 Element: 20524

Model name: kh
Study name: Disc Brake Static(-Default-)
Plot type: Static strain Strain1
Deformation scale: 296.135



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Disc Brake Static-Strain

